Recommender Systems

From Foundations to Frontiers

Lê Thiện Hòa

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01 **Foundations** — Core principles of personalization: users, items, interactions, evaluation

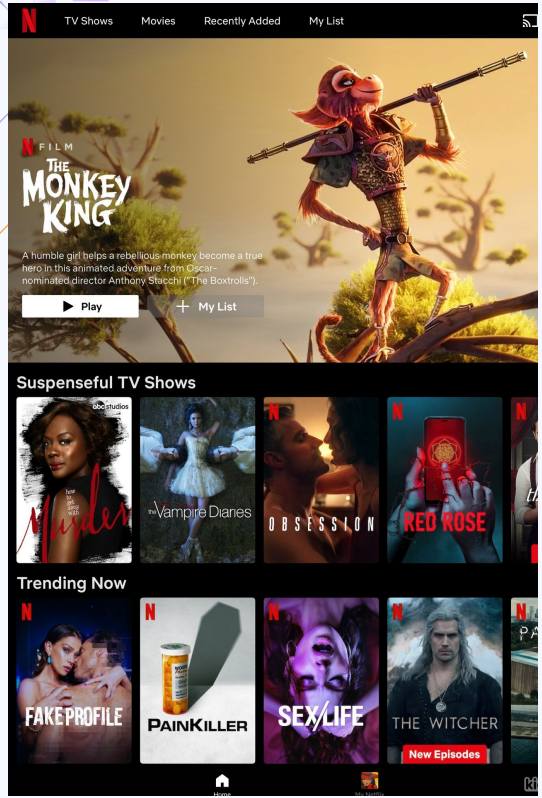
02 **RecSys Models** — From matrix factorization to deep learning, powering recommendations

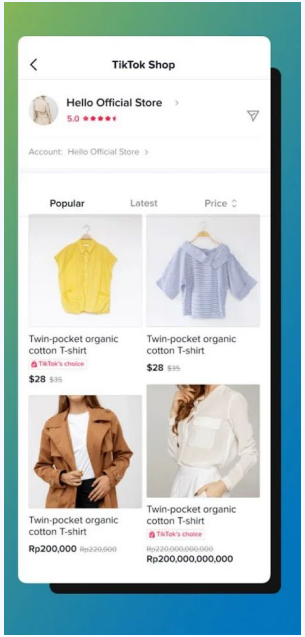
03 **Beyond** — Future of RecSys

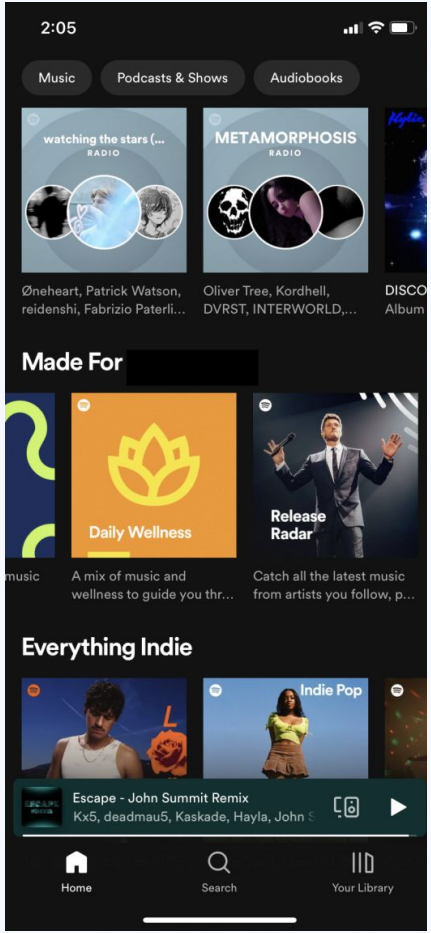
01

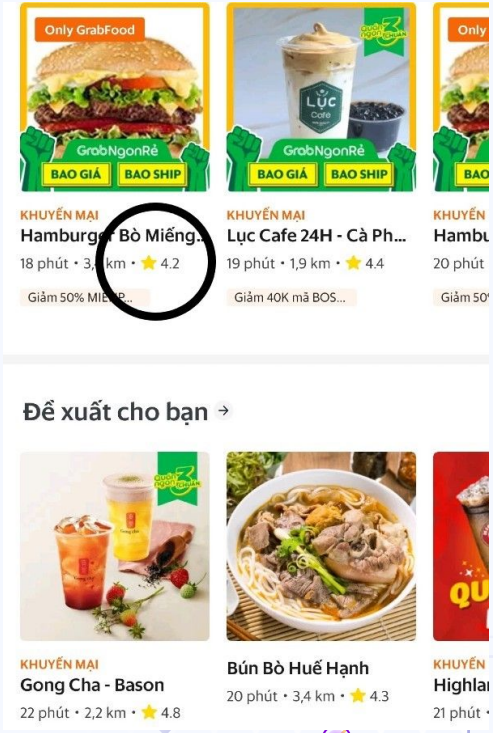
Foundations

Recommendation everywhere









Collaborative filtering

Watched by both women (W1 and W2)



Similar users





Watched by W1



Recommended to W2

Content-based filtering

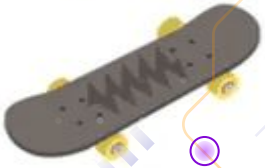
Towards Exploitation

Watched by user



Similar items



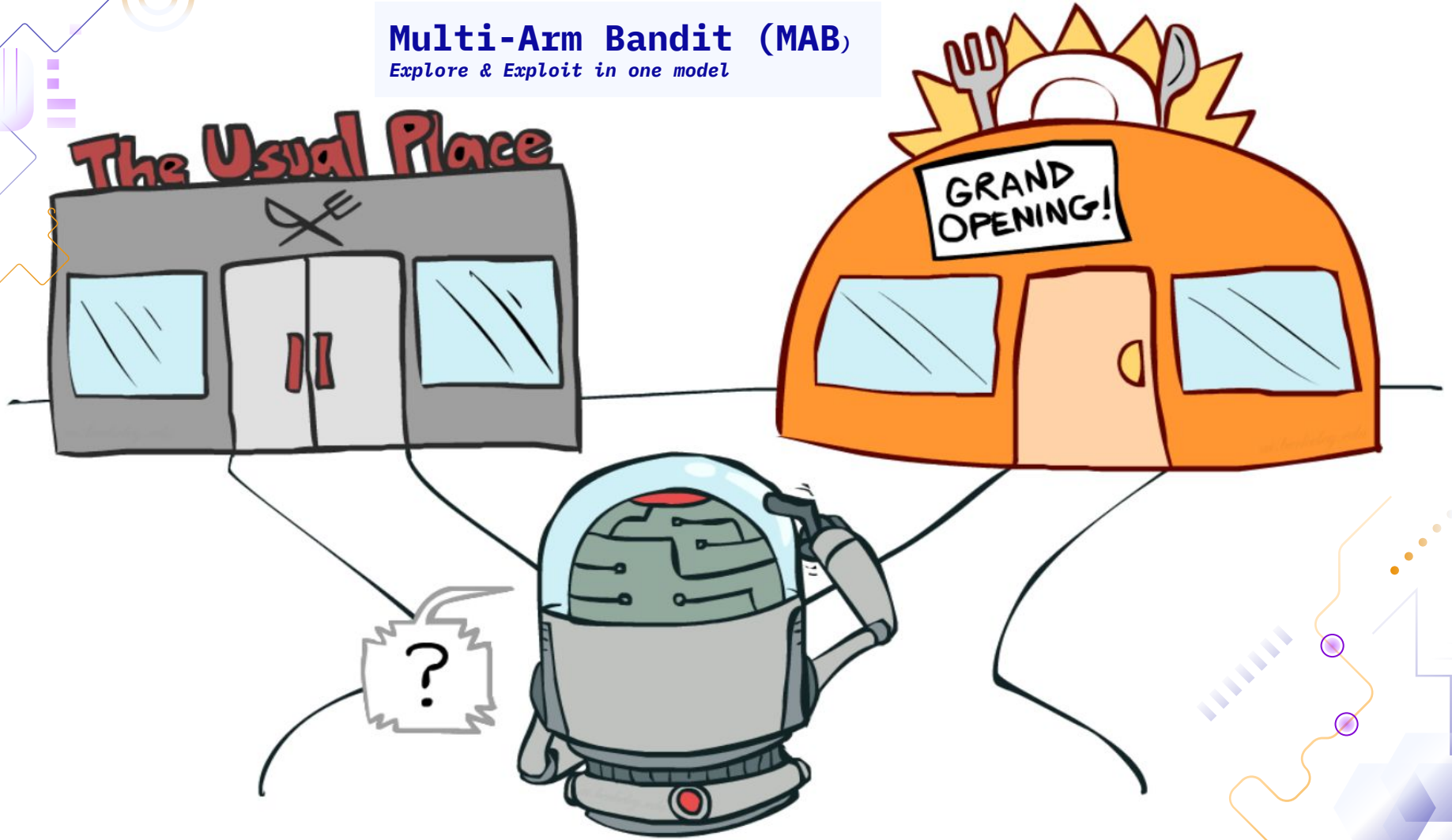


Recommended to user

Towards Exploration

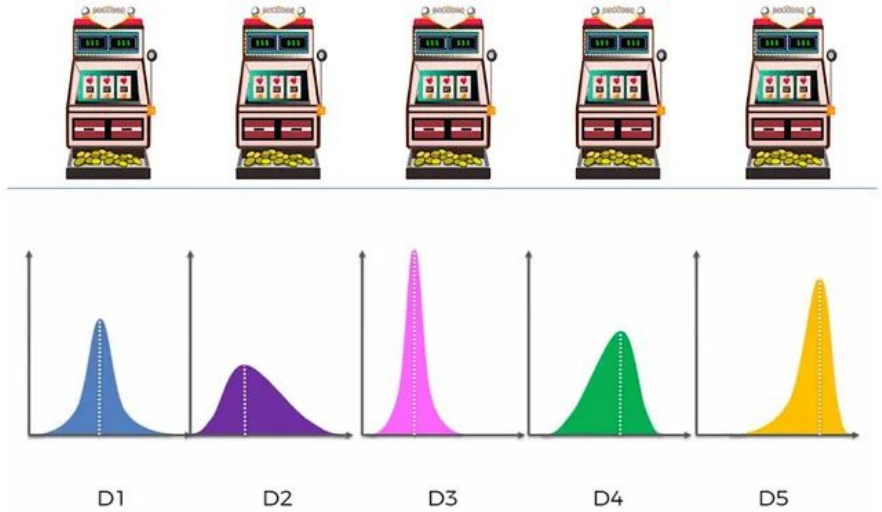
Multi-Arm Bandit (MAB)

Explore & Exploit in one model



Multi-Arm Bandit (MAB)

Explore & Exploit in one model



- **Scenario:** Pull machine k → sample from ***unknown*** reward distribution D_k → observe reward.
- **Problem:** Given a finite number of pulls T, how can I optimize my winnings?
- How much should I **explore**? How much should I **exploit**?

Evaluation

1. Rating Prediction (Regression)

- RMSE
- MAE

2. Ranking Quality (Top-N Recommendation)

- Precision@K
- Recall@K
- F1@K
- NDCG
- Hit Rate@K

3. Coverage & Diversity

- Catalog Coverage
- User Coverage
- Diversity
- Novelty / Serendipity

4. Business / System Metrics

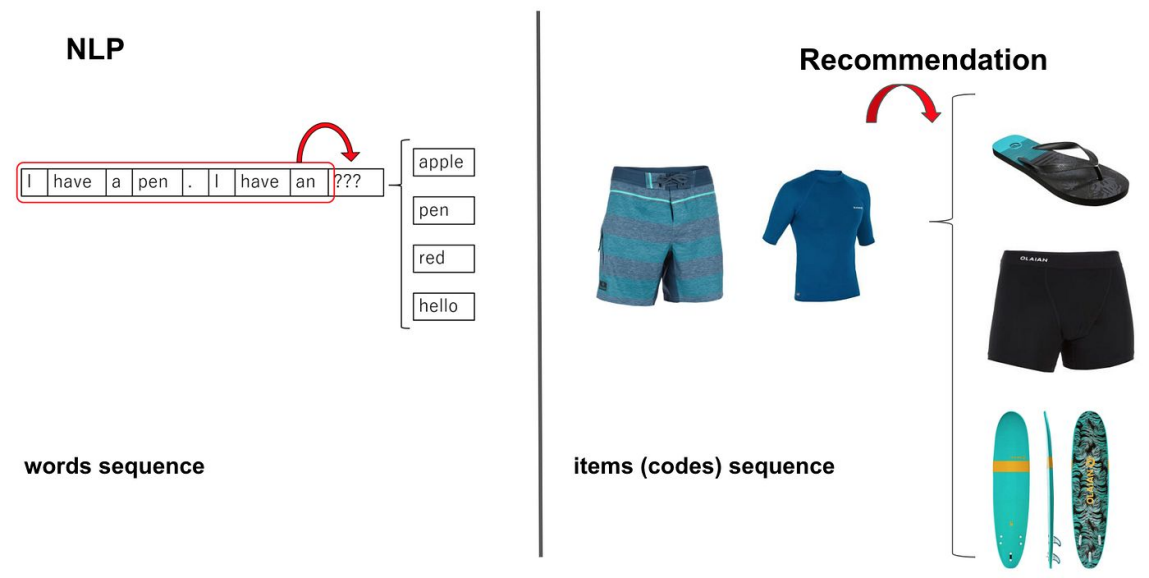
- CTR
- Conversion Rate
- Latency & Scalability

02

Models

1D - Word2vec

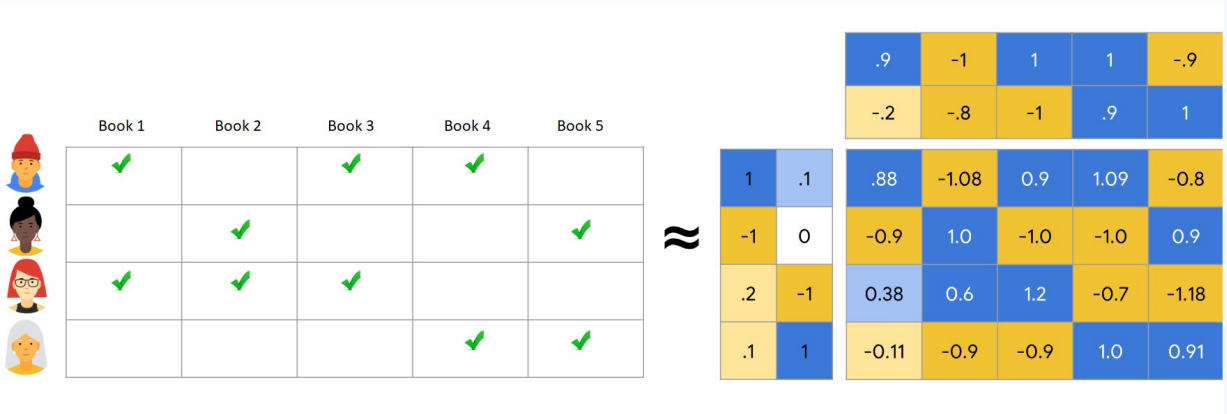
Co-Occurrence (without Personalization)



- Learns embeddings only for **items** (or only users).
- Example: **Word2Vec** / **Item2Vec** – items embedded from user behavior sequences, *no explicit user embedding*.

2D - Collaborative Filtering

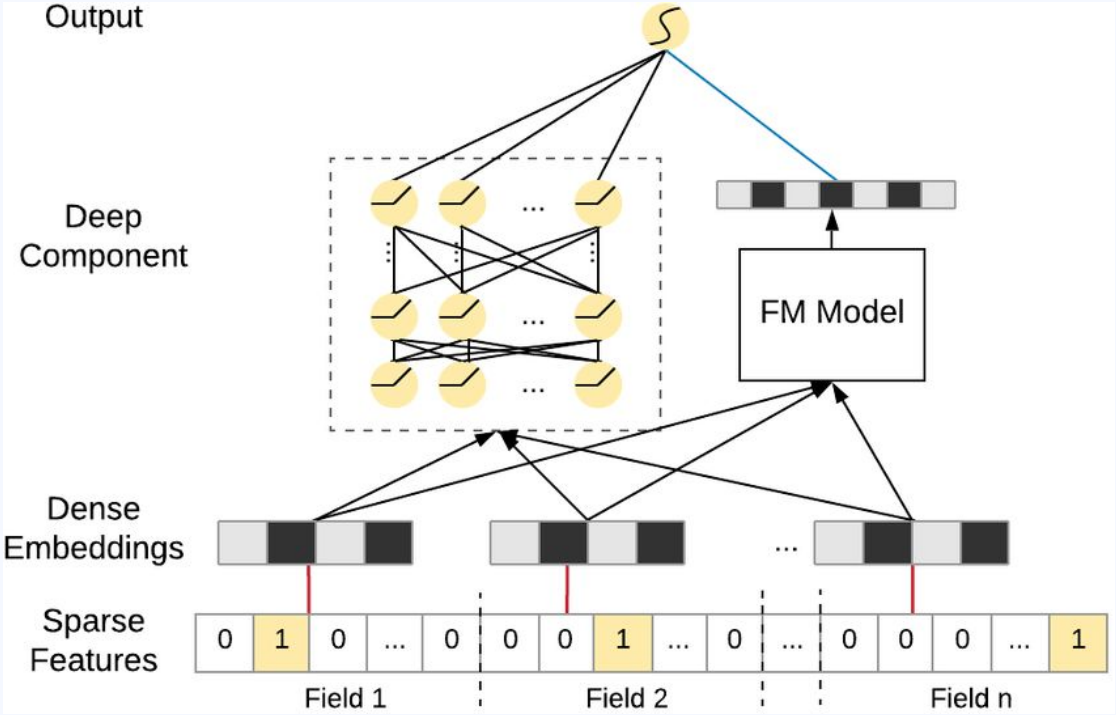
Matrix Factorization (Ex: ALS)



- Learns **embeddings** for both **users** and **items**.
- Interaction modeled as **dot product** or similarity between user and item vectors.

Multi-D - Hybrid RecSys (Ex: DeepFM)

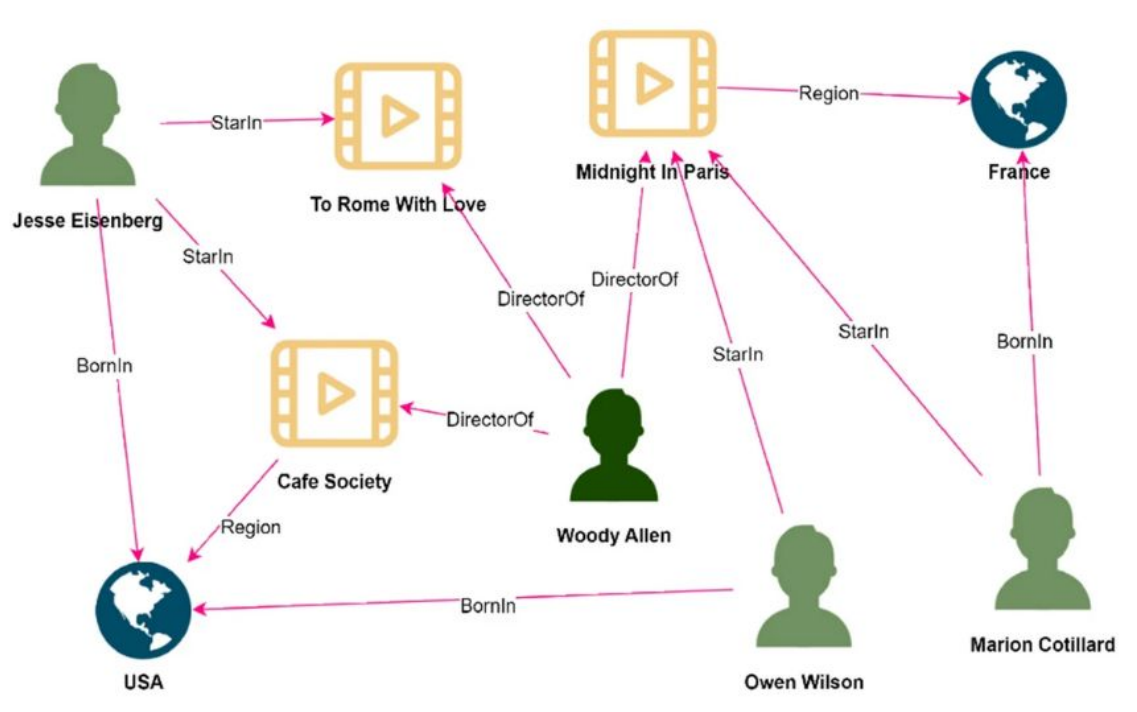
Content-based + Collaborative Filtering



- Incorporates **user**, **item**, and **additional features** (context, location, side-info, social/graph signals).
- Learns embeddings for multiple fields and models **higher-order interactions**.

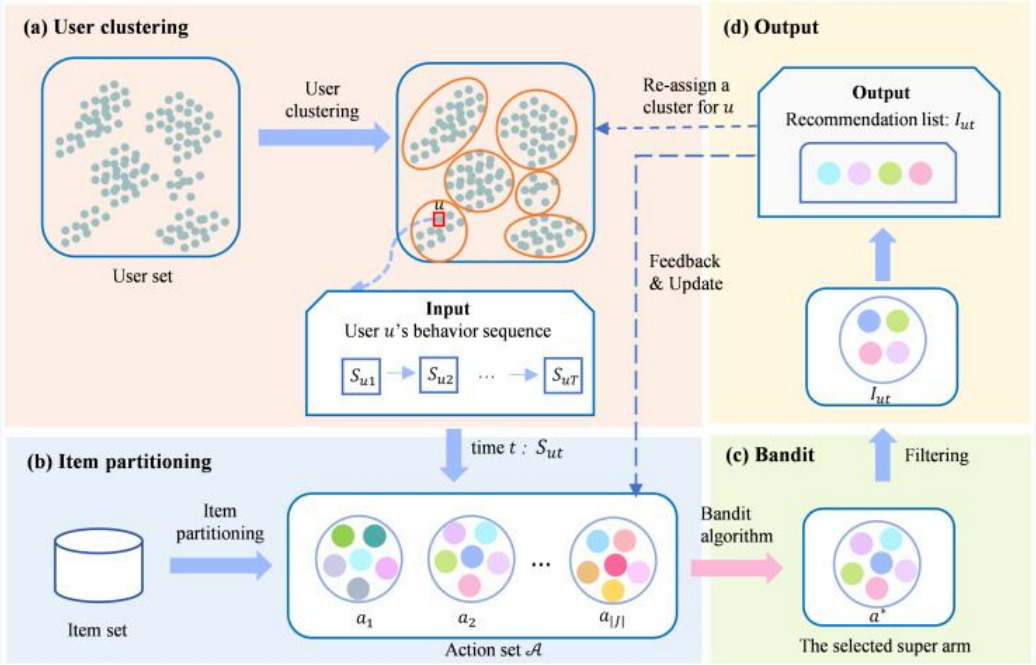
Multi-D - Knowledge Graph RecSys

Ex: GraphSage, Metapath2vec



- Represents users, items, and their relationships as interconnected nodes.
- Graph connections are more **compact** and **expressive** than tabular data.
- Learns embeddings by aggregating information from neighbors and edges.
- Captures higher-order, **multi-hop interactions** beyond direct user–item links.

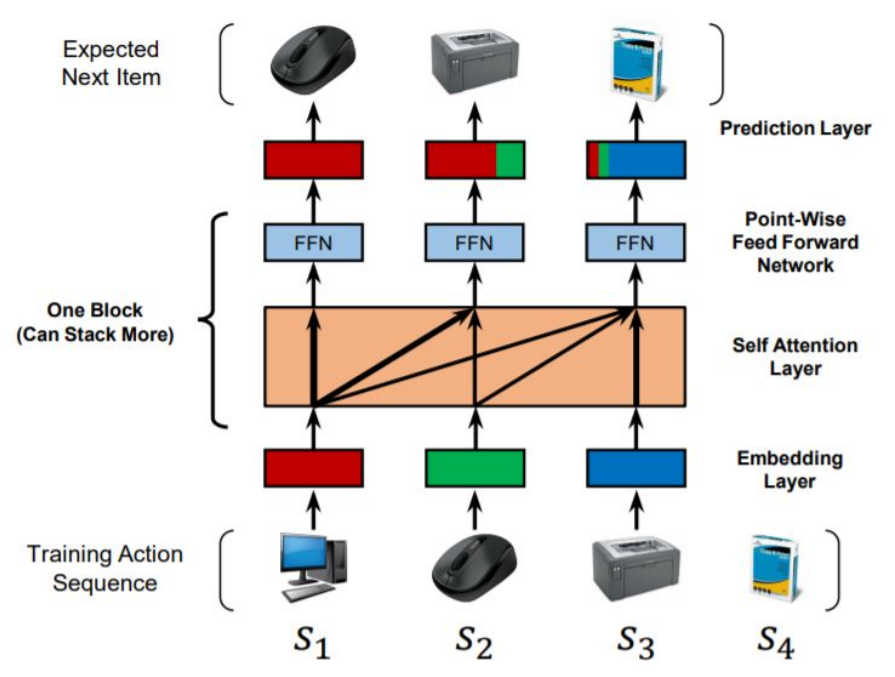
Multi-D - Clustered (& Contextual) MAB



- Groups similar users into clusters.
- Each cluster shares bandit policy → more data per arm.
- Helps converge faster by pooling experience within clusters.
- Balances personalization vs. learning speed

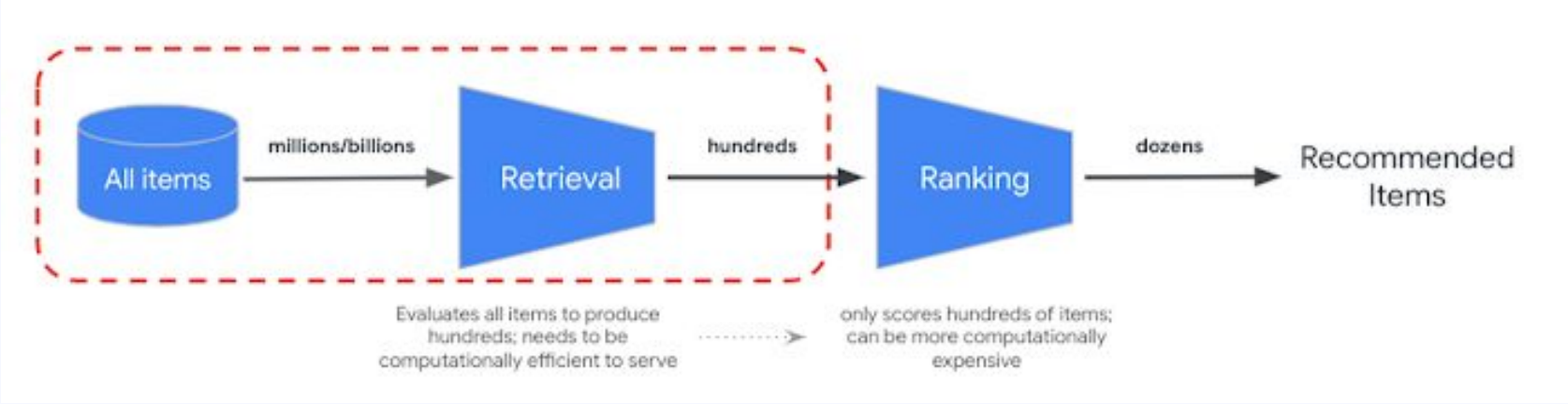
Time-D - Sequential Recommendation

Ex: SASRec, Bert4Rec



- User behavior is inherently **dependent**, current action depends on both **user interests** and **past actions**
- Models sequences using **Transformers**, or attention mechanisms
- Captures **evolving preferences** and **dynamic short-term user intent**

Recommendation Systems



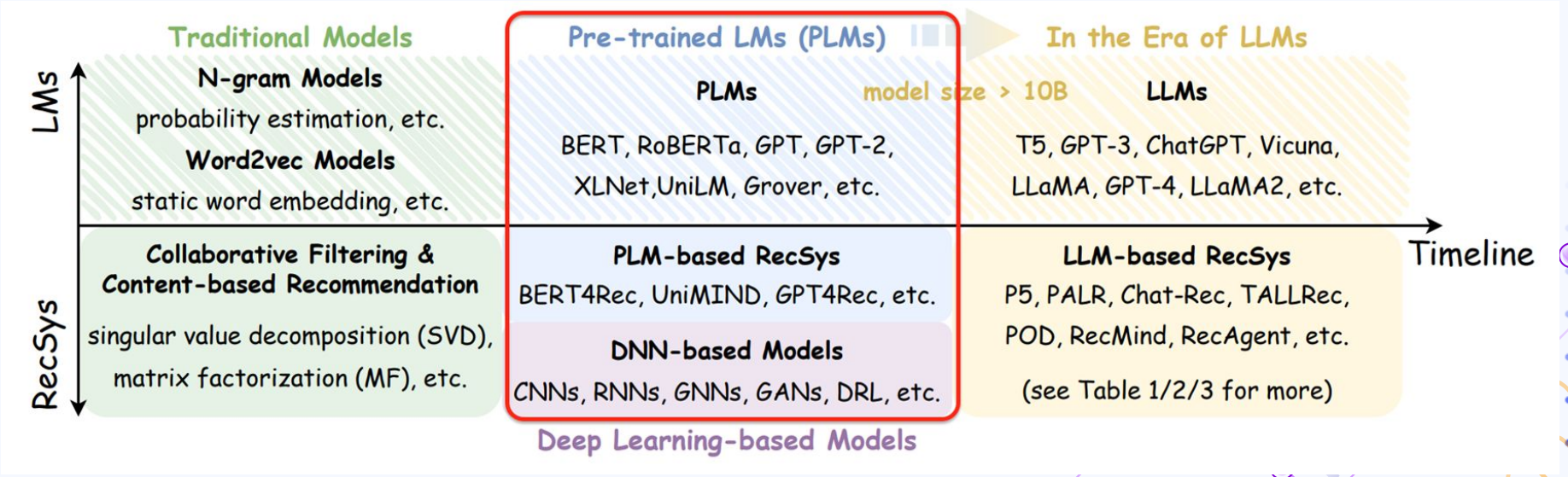
- **Objective**: Diversify & Explore
- **Metric**: Recall
- **Models**: Matrix Factorization
  - CF (ex: ALS)
  - Word2vec
  - GraphSAGE

- **Focus**: Relevant
- **Metric**: Precision
- **Models**: Deep Learning
  - Content-based & CF (DeepFM)
  - Sequential RecSys (Bert4Rec)

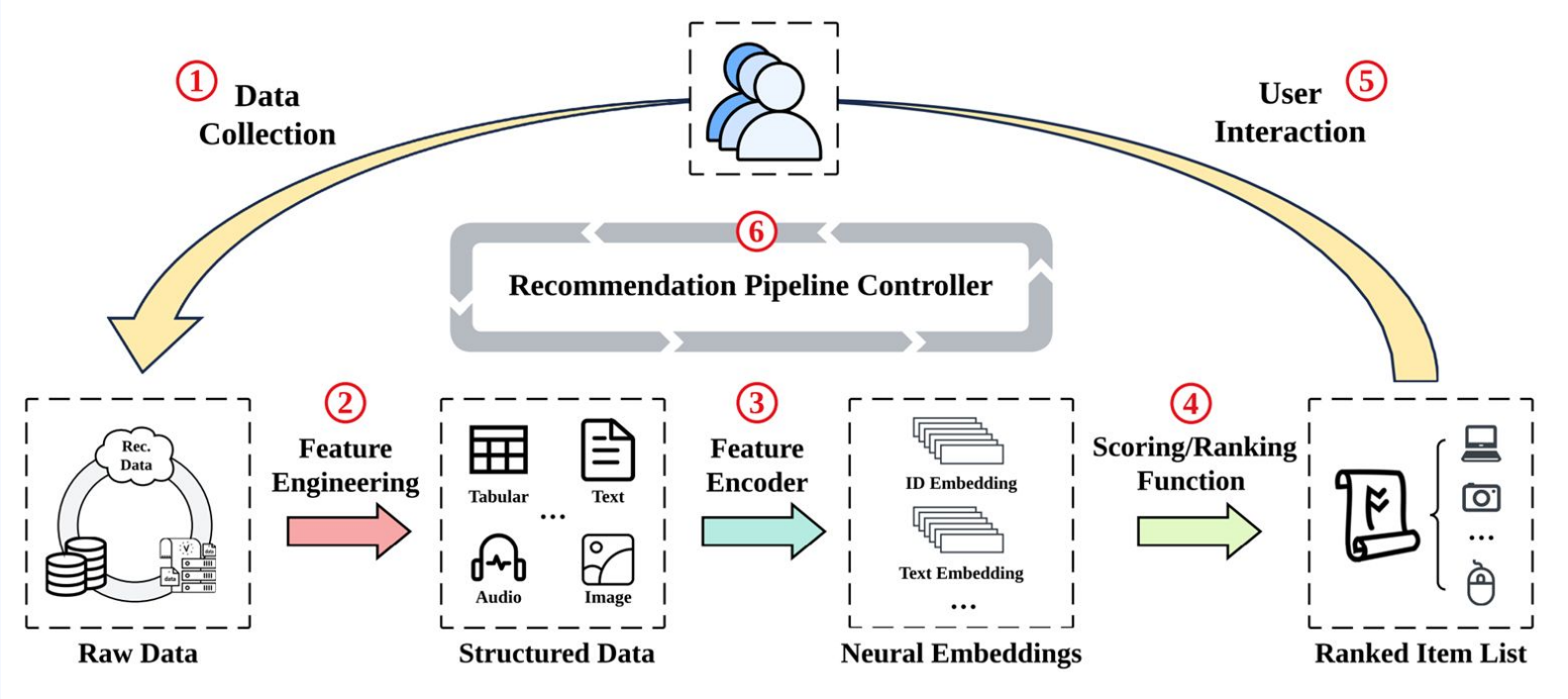
03

Beyond RecSys

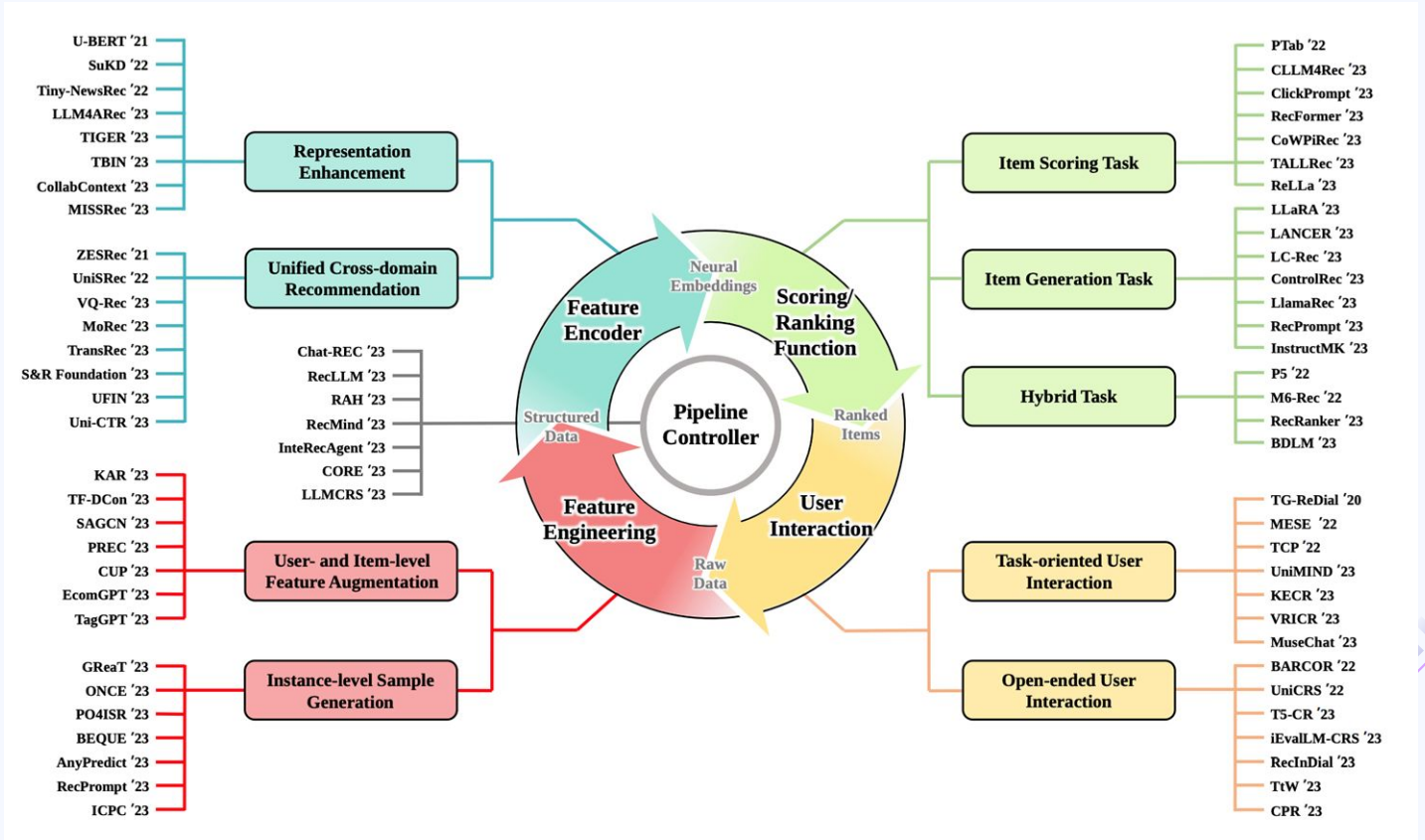
Evolution of LLM & RecSys



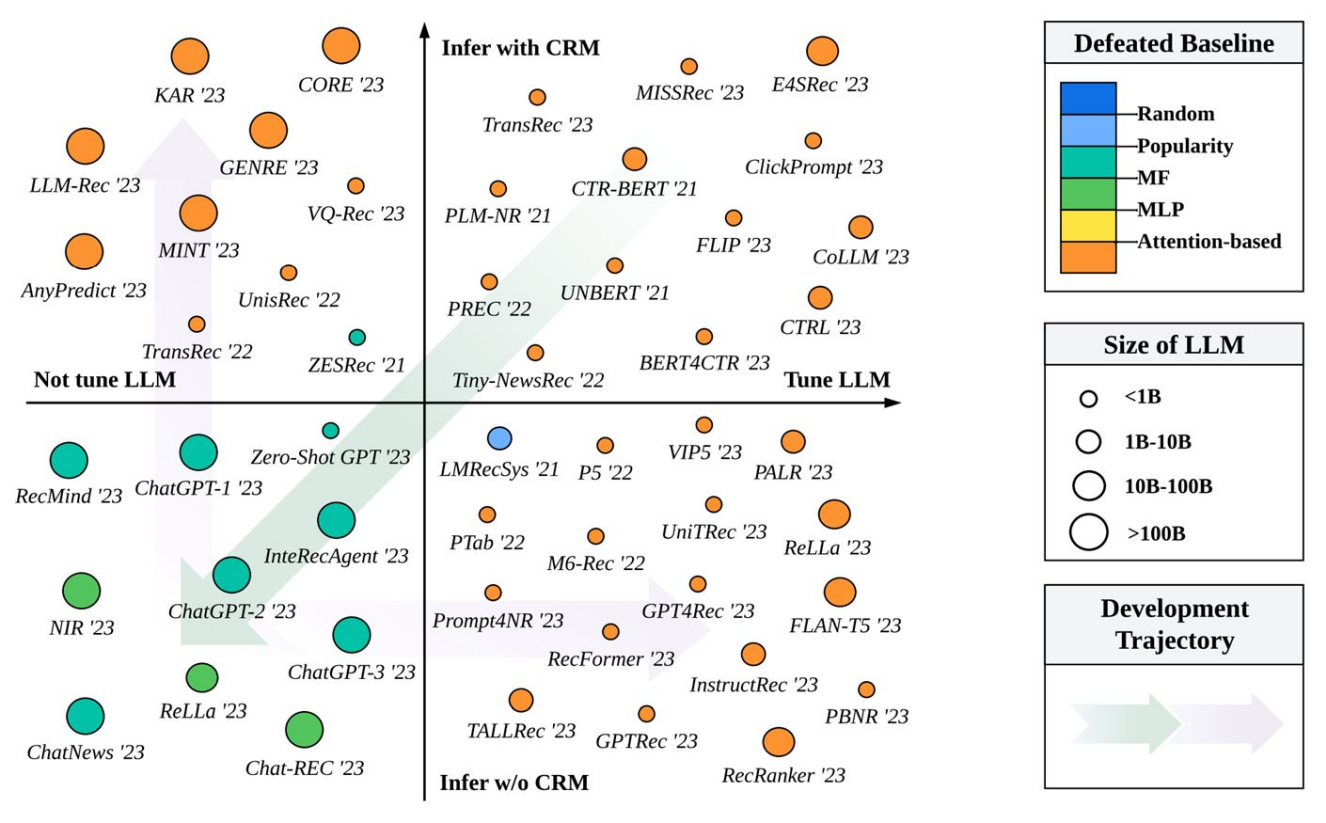
A typical DL based RecSys pipeline



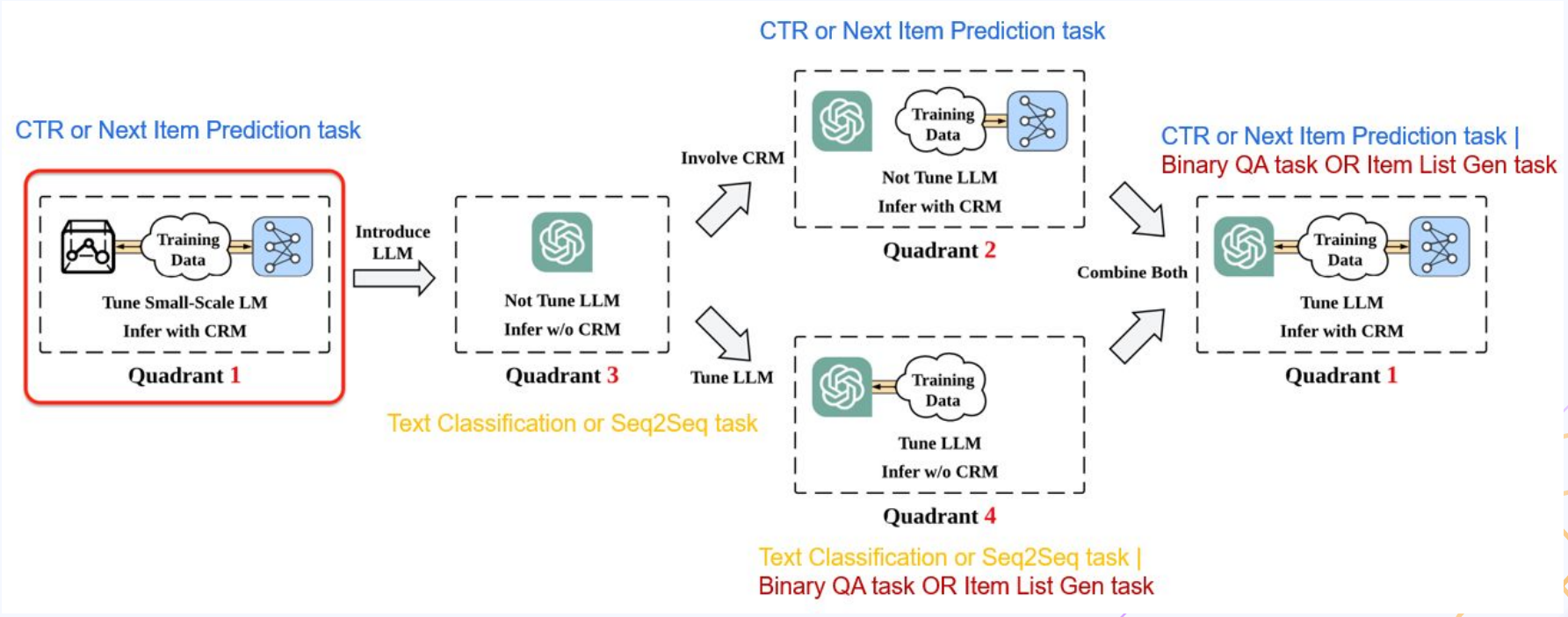
LLM for RecSys



Evolution of LLM (tune or not) with RecSys (infer with / without)



Evolution of LLM (tune or not) with RecSys (infer with / without)



Summary: Challenges of using LLM for Real World Applications

- Efficiency
  - Training efficiency: LORA finetuned efficiently on Llama 7B, GPU RTX 3k
  - Inference Latency: PEFT only solve memory usage problem
    - **Pre-computing and caching**
    - **Reduce the size of the model:** Distillation, Pruning, Quantization
- Effectiveness
  - Long Text Modeling: **filter, select, and arrange the key textual information**
  - ID Indexing & Modeling: currently both works, best approach is still an open research problem
- Risks
  - Fairness: user bias (gender, age, race,...) & item bias
  - Hallucination
  - Privacy
  - Explainability

Resources:

- Exploring the Impact of Large Language Models on Recommender Systems: An Extensive Review, arxiv 2024, [paper]
- A Survey on Large Language Models for Recommendation, arxiv 2023, [paper]
- How Can Recommender Systems Benefit from Large Language Models: A Survey, arxiv 2023, [paper]
- Recommender Systems in the Era of Large Language Models (LLMs), arxiv 2023, [paper]